

Session 1
Start of the Lesson (10 minutes)
Answer the prompts below on your own, then share your ideas with your group. As a group, choose one strong insight to share with the class.
1. How does the Design Thinking process guide the way your team developed and improved your prototype? <u>The Design Thinking process helped our team understand user needs, identify problems, generate ideas, and improve the prototype through testing and feedback.</u>
2. What ethical or user-centered principle has most influenced the design and functionality of your prototype so far? <u>The most important principle is providing accurate and reliable supplement information to ensure user safety and prevent misinformation.</u>
3. How did your problem statement, project scope, or WBS help shape the decisions you made during your prototype build? <u>It helped the team focus on important features, organize tasks, and avoid unnecessary functions during development.</u>
4. Which tools, technologies, or prototype features best represent your solution — and what kind of research evidence or rationale supports these choices? <u>React.js, Node.js, Express.js, MongoDB, and AI-based recommendation features best represent our solution because they support fast web development, personalized recommendations, and efficient data management.</u>

Main Task – Try It Out: Why Does It Work?

1. Why is it important to justify the choices made in your prototype (tools, design, structure)?

It is important because justification explains why the selected tools and designs are suitable for solving the identified problem and meeting user needs.

2. What could happen if your design or technology decisions aren't supported by research or evidence?

The system may become ineffective, unreliable, difficult to use, or unable to solve the actual problem of users.

3. How do user needs and project goals influence the tools and methods you select?

User needs and project goals help determine the most appropriate technologies, features, and methods to ensure the system is useful and functional.

Let's Practice: Research Brief & Technical Rationale (45 minutes)

Step 2: Draft Your Research Brief

Research Brief Template

Project Title	SuppleWise: A Web-Based Dietary Supplement Guidance System Using Artificial Intelligence
Problem Context	studies have shown that many consumers rely on unreliable sources when selecting supplements, leading to misuse and potential health risks
Users	Users, Patients, and health care professionals.
Evidence of Need	Kantar et al., 2021
Prototype Link/Description	We build a HomePage, login and registration page, and supplement recommendations using the MERN stack.
Solution Viability	AI technology can analyze user data and generate personalized supplement recommendations while improving accessibility and convenience.

Step 3: Complete the Tech Rationale Table

Tech Rationale Table

Category	Tool/Technology	Why Chosen	Alternatives Considered	Link to User Needs/Prototype
Design	Figma	Easy to use for wireframes and UI design	Canva, Adobe XD	Helps visualize the system clearly
Frontend	React.js	Creates responsive and interactive interfaces	Html, css, JavaScript	Improves user experience and accessibility
Backend	Node.js & Express.js	Efficient for server-side functions and	Laravel, PHP	Supports fast system processing

		APIs		
Database	MongoDB	Flexible and scalable database	MySQL	Stores user and supplement data efficiently
Other Tools	GitHub & VS Code	Collaboration and coding support	GitLab, Sublime Text	Helps manage development

Session 2
Start of the Lesson (10 minutes)
For each statement below, write True (T) if you agree or False (F) if you disagree — then explain why in one short sentence. Work individually first, then discuss your answers with your group. 1. A Research Brief only lists what you built; it doesn't need to explain the evidence or reasoning behind your solution. False – A Research Brief also includes user needs, evidence, and proposed solutions. 2. A Technical Rationale should justify tool and technology choices using research, user needs, and project goals. True – A Technical Rationale explains why the selected tools and technologies are appropriate for the project. 3. It's acceptable for teams to use different design tools (like Figma, Canva, Penpot, or Adobe XD) as long as they can explain why the tool fits their project. True – Different tools can be used if they effectively support the project's needs and goals. 4. Linking your prototype decisions to earlier modules (like user needs, ethics, and scope) helps show that your design process is well-founded and intentional. True – Connecting prototype decisions to earlier modules shows that the design process is organized and research-based.

Main Task – Try It Out: Feedback Matters

1. What is one part of your Research Brief you're most confident about?

The problem context because it clearly explains the issue of misinformation regarding dietary supplements.

2. What is one area in your Technical Brief that might still need clearer justification?

The AI recommendation feature may still need stronger justification and testing evidence.

3. How can peer feedback help your group improve the documentation or prototype?

Peer feedback helps identify weaknesses, improve system features, and make the documentation clearer and more organized.

Think Back and Think About (10 minutes)

Think Back: How did creating your Research Brief and Technical Rationale help your team connect your prototype decisions to research evidence, user needs, and design principles?

The research brief and technical rationale helped our team connect prototype decisions to research evidence, user needs, and project goals. It also guided us in choosing suitable tools and technologies for the system.

Think About: What valuable insights did your team gain from the **peer presentation and feedback session**, and how will you apply them to **improve or refine** your Capstone Project in the next phase

We learned the importance of research, teamwork, communication, and proper documentation while developing the project. These lessons will help us improve the prototype, solve problems effectively, and continue developing our Capstone Project successfully.